

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks: 40

Annexure A

CASE STUDY

Code of Conduct: (192565062)

I K. Reshma (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Reshma

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89

10

K. Ruhma

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CSA1714

(03 AT)

Artificial

Intelligent

(a) Propositional Logic Representation

Let,

F = patient has Fever

C = patient has cough

R = patient has Rash

B = patient has Breathless

Flu = patient has possible Flu.

M = patient has possible measles

P = patient has possible pneumonia

$\wedge$ = AND

$\rightarrow$ = Implies

Rules

1. Fever AND

$$F \wedge C \rightarrow \text{Flu}$$

Cough $\rightarrow$ possible Flu

2. Fever AND

$$F \wedge R \rightarrow M$$

Rash $\rightarrow$ possible Mumps

3. cough AND

$$C \wedge B \rightarrow P$$

Breakleaves $\rightarrow$ possible pneumonia

Facts about patient A

$$F = \text{True}$$

$$C = \text{True}$$

$$B = \text{True}$$

(b) forward chaining

Initial Facts:

F, C, B

Step 1

Rule 1:

$$F \wedge C \rightarrow \text{Flu}$$

since $F = \text{True}$ and $C = \text{True}$

$Iu = \text{True}$

Rule 2:

$F \wedge R \rightarrow M$

we have $F = \text{True}$, but R is not given.

Therefore, measles cannot be derived.

step 3:

$C \wedge B \rightarrow P$

since $C = \text{True}$ and $B = \text{True}$

pneumonia = True

End Diagnoses

possible Flu

possible pneumonia

measles cannot be concluded because R is not given.

(c) Backward chaining

Goals:

pneumonia

Rule 3:

cough $\wedge$ Breathlessness

so the goal is reduced to:

Goal: pneumonia

cough AND Breathlessness

cough
Breathlessness

Both facts are already present for patient A:

cough = True

Breathlessness = True

Therefore:

patient A has possible pneumonia.

Result: The goal is successfully reached

using Backward chaining.

one Reaction

Rule 1:

$\text{vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{clearPath}$.

Given:

$\text{vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

so,

$$\Theta = \{x \mid \text{Ambulance}\}$$

Therefore:

$\text{clearPath}(\text{Ambulance})$

by Resolution

Rule 1:

$\text{vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{clearPath}$

using:

$\text{vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})$

we get:

$\text{clearPath}(\text{Ambulance})$

Rule 2:

$\neg \text{clearPath}(x) \vee \text{GreenSignal}(x)$

Therefore:

GreenSignal (Ambulance)

Rule 3:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(\text{Ambulance})$

Given:

$\text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

Substitute:

$x = \text{CarA}, y = \text{Ambulance}$

Therefore:

$\text{Proceed}(\text{CarA})$

So, $\text{proceed}(\text{CarA})$ is true.

CC) Two Emergency vehicles

No, the current knowledge base is not sufficient
we need additional facts such as:

$\text{Vehicle}(\text{FireTruck})$

$\text{EmergencyType}(\text{FireTruck})$

And rules for:

- * Vehicle priority.
- * Route conflicts
- * Traffic signal control.

soil Dry $\rightarrow$ Irrigation Needed.

7 Soil Dry v Irrigation Needed.

P2:

Irrigation Needed ^ CropWheat $\rightarrow$ Apply Drip

Irrigation Needed v CropWheat ^ Apply Drip.

P3:

ApplyDripMethod ^ CropWheat $\rightarrow$ CropAtRisk

ApplyDripMethod v CropWheat v CropAtRisk.

Facts:

soilDry, cropWheat

(6) Resolution Refutation

Goals:

ApplyDripMethod

negate:

ApplyDripMethod.

Step 1:

soilDry $\rightarrow$ Irrigation Needed.

So,

Irrigation Needed

Step 2:

Irrigation Needed + Crop wheat

Apply Drip Method.

Step 3:

Apply Drip Method + 2 Apply Drip Method.

Therefore,

Apply Drip Method is TRUE.

(C) If soil Dry is Removed

Without soil Dry:

~~Irrigation Needed~~ → cannot derive.

~~Apply Drip Method~~ → cannot derive.

~~Crop At Risk~~ → cannot derive.

Crop At Risk does NOT follow.

Find: soil Dry is necessary to derive

Irrigation Needed and Apply Drip Method.

percept state + Modus ponens

perceptions:

stench at (1,2)

Breeze at (1,1)

Gletter at (2,2)

Apply Modus ponens

step 1:

stench(1,2) $\rightarrow$ wumpus is adjacent to (1,2)

Therefore:

wumpus is adjacent to (1,2)

step 2:

Breeze(1,1) $\rightarrow$ pet is adjacent to (1,1)

Therefore:

pet is adjacent to (1,1)

step 3:

Gletter(2,2) $\rightarrow$ Gold is at (2,2)

Therefore:

Gold is at (2,2)

Derived Facts:

Wumpus is adjacent to $[1,2]$

Pet is adjacent to $[1,1]$

Gold is at $[2,2]$

(b) Forward chaining

Wumpus

Stench at $[1,2]$ means Wumpus is in a cell

adjacent to $[1,2]$

possible cells:

$[1,1], [1,3], [2,2]$

Pet

Breeze at $[1,1]$ means Pet is in a cell

adjacent to $[1,1]$

possible cells:

$[1,2], [2,1]$

the path safe?

$(1,1) \rightarrow (1,2) \rightarrow (2,2)$

we know?

pet may be at $(1,2)$

wumpus may be adjacent to $(1,2)$

Gold is at $(2,2)$

Therefore, the path cannot be guaranteed safe.

path is NOT confirmed safe.

conclusion:

The agent should not assume the path is safe because $(1,2)$ may contain a pet and there may be a wumpus nearby.